

MIEFormer: Multi-band Iterative Refinement via Progressive Residual Learning for Low-Light Enhancement

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Abstract—Enhancing night vision images under extremely low illuminance (below 10^{-3} lux) faces three fundamental challenges: the absence of dedicated datasets covering this extreme illumination range, the scarcity of recoverable signal information due to ultra-low signal-to-noise ratios, and the difficulty of balancing efficient feature learning with robust generalization. To address these challenges, we first construct the Enhanced Low-light Night Vision Dataset (ELND), a multi-scene benchmark spanning indoor illuminance from 1×10^{-1} to 3×10^{-4} lux and outdoor scenarios including moonlight, starlight, and complete darkness. We then propose MIEFormer, a multi-band iterative enhancement framework that decomposes the illumination spectrum into discrete frequency bands and progressively refines features via residual learning stages. Specifically, its core module, the Multifrequency and Content-guided Attention (MFCA) module, which combines an Adaptive Band-pass Filtering Attention (ABFAttention) for global multi-frequency feature fusion and a Multibranch Content-Guided Convolution (MCGConv) for adaptive local refinement. Extensive experiments conducted on the ELND and public LOL datasets demonstrate that MIEFormer significantly outperforms existing state-of-the-art methods on the extreme low-light ELND dataset, while achieving competitive performance on the conventional low-light LOL dataset. Specifically, it achieves 21.29 dB in PSNR and 0.66 in SSIM on the indoor 3×10^{-4} lux sequence of ELND, and 22.68 dB in PSNR and 0.73 in SSIM on the outdoor starlight scenario of ELND, thus establishing a new benchmark for extreme low-light enhancement.

Index Terms—Low-Light Enhancement, Extremely Low-illumination, Enhanced Low-light Dataset, Progressive Residual Learning

I. INTRODUCTION

LOW-LIGHT night vision imaging converts scenes beyond human perception into recognizable images through photoelectric conversion and signal enhancement [1]–[3]. As a passive sensing modality, it ensures high concealment and consistent image fidelity, enabling widespread deployment in autonomous driving, surveillance, remote sensing, and industrial inspection [1], [2]. Despite recent advances in CMOS sensors and optical systems, extremely low illumination (e.g., 10^{-3} – 10^{-4} lux) still poses critical challenges: severely degraded signal-to-noise ratio (SNR), imbalanced global-local

contrast, amplified noise, and loss of fine details [4], [5]. Consequently, enhancing night vision under such demanding conditions remains an open and significant problem.

Conventional enhancement methods predominantly adopt two strategies: extending exposure time or increasing sensor gain [2], [5], [6]. Prolonged exposure integrates more incident photons but inevitably reduces temporal resolution, rendering it unsuitable for dynamic scenes with moving objects [5], [6]. Increasing gain amplifies per-pixel sensitivity to faint illumination [5], yet it indiscriminately boosts signal and noise, leading to severe SNR degradation and substantially compromising image quality for downstream tasks such as detection, recognition, and semantic understanding [7]–[10].

In contrast to hardware-based adjustments, data-driven approaches, especially deep learning models, have recently shown considerable promise for low-light image enhancement [8], [9]. Instead of altering acquisition parameters, these methods learn direct mappings from low-light inputs to well-exposed counterparts, improving contrast, suppressing noise, and recovering lost details in an end-to-end fashion [1], [7]–[9]. However, despite their success under typical low-light settings, existing deep learning models remain inadequate for extremely low-illumination night vision scenarios, primarily due to three critical challenges:

1) Limited illumination coverage in existing datasets. Illumination conditions in most public datasets (e.g., SID [11], SDS [12], and SMID [6]) are not strictly quantified due to uncontrolled capture settings, which limit their ability to represent real-world extreme low-light environments. Moreover, data collection is often restricted to specific periods (e.g., dusk or dawn) and scenes, failing to include conditions like moonless starlight. As a result, available data are largely confined to the range of 10^{-1} – 10^{-2} lux, while the critically important regime below 10^{-3} lux remains markedly underrepresented.

2) Severe degradation of effective information. Under extremely low-light conditions, imaging sensors capture only a sparse stream of incident photons, leading to substantial deterioration in both brightness and fine detail. In terms of brightness, the dynamic range of captured images is heavily compressed toward the lower end, confining most pixels to a narrow low-intensity interval. Regarding fine details, the vital high-frequency information that carries essential texture and edge features is largely overwhelmed by intense noise, leading to severe degradation or even complete loss.

3) Inherent limitations in model architecture. Current deep learning methods predominantly rely on CNN or Vision Transformer backbones. While CNNs effectively capture local patterns, they struggle to model long-range dependencies,

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which are crucial for handling non-uniform illumination and performing global contrast adjustment [1], [7]–[9]. Although CNN-based approaches can be enhanced via multi-branch designs or frequency-domain operations, their ability to incorporate global context remains fundamentally constrained. Vision Transformers, by contrast, excel at capturing global relationships but suffer from high computational complexity, hindering deployment in real-time or resource-constrained applications [13]. Although window-based and sparse attention mechanisms reduce computational cost, they often weaken the model’s capacity to leverage full-image context, which limits their effectiveness in challenging low-light scenarios. Although window-based or sparse attention reduces computation, they often weaken the model’s capacity to leverage full-image context, thus limiting effectiveness in challenging low-light scenarios [1], [7]–[9], [14].

To address these challenges, we first construct the Enhanced Low-light Night Dataset (ELND), a multi-illumination benchmark specifically designed for extreme low-light enhancement. Its collection and processing follow three principles to ensure accuracy and practical relevance:

- 1) Precisely Controlled Indoor Acquisition. Images are captured in a professional optical darkroom using calibrated devices at five strictly quantified illumination levels: 10^{-1} , 10^{-2} , 10^{-3} , 3×10^{-3} , and 3×10^{-4} lux, guaranteeing consistent and measurable lighting conditions.
- 2) High-quality Reference Generation. By incorporating calibrated sensor parameters, we produce corresponding high-quality reference images using professional post-processing tools, serving as reliable ground truth for objective evaluation.
- 3) Diverse Outdoor Generalization Test Set. To assess real-world applicability, we include an outdoor test set captured under three natural low-light scenarios: moonlit, starlit, and overcast (no direct light) nights, enhancing the benchmark’s practical relevance and diversity.

The illumination intervals defined in ELND naturally motivate a piecewise enhancement strategy, as image degradation characteristics vary sharply across different lux levels. Building on this observation, we formulate the enhancement task as a piecewise residual learning process across multiple illumination intervals. Specifically, motivated by an empirical analysis of frequency-domain amplitude variations and signal-to-noise ratio (SNR) at different light levels (see Section III), we first discretize the continuous illumination spectrum from extreme low-light to normal light into distinct intervals. Within each interval, we apply supervised residual learning that jointly leverages frequency- and spatial-domain representations: the frequency domain captures global degradation patterns, while the spatial domain refines local structures. These intervals are processed iteratively to progressively enhance the image until the target quality is achieved.

Following this framework, we propose the Multi-band Iterative Enhancement Former (MIEFormer). For each illumination interval, MIEFormer first decomposes the input into multiple frequency bands via a bank of multi-scale Gaussian band-pass filters. A frequency-aware fusion module then inte-

grates these bands to strengthen global contextual modeling. Complementarily, a Multi-Branch Content-Guided Convolution (MCGConv) module is designed to enhance local feature representation and generalization. The model is trained in a supervised manner to perform interval-specific enhancement. Experiments demonstrate that MIEFormer achieves state-of-the-art performance on the proposed benchmark.

The main contributions of this paper are summarized as follows:

- 1) We introduce the Enhanced Low-light Night Vision Dataset (ELND), a novel benchmark with precisely quantified illumination levels for extreme low-light night vision research. Acquired using specialized optical equipment, it spans a wide range from 10^{-1} to 3×10^{-4} lux and includes a diverse outdoor test set, providing a valuable resource for the community.
- 2) We propose a piecewise iterative enhancement framework that reformulates the overall task into successive refinements across discrete illumination intervals. Based on this, we develop the MIEFormer network.
- 3) We design the Multi-frequency and Content-guided Attention (MFCA) module, which synergizes an Adaptive Band-pass Filtering Attention (ABFAttention) for global multi-frequency context with a Multi-branch Content-Guided Convolution (MCGConv) for local feature refinement.
- 4) Extensive experiments on the ELND and LOL datasets demonstrate the superiority of the proposed method, demonstrating significant improvements in both quantitative metrics and subjective visual quality under extremely low-light conditions.

The remainder of this paper is organized as follows: Firstly, we briefly review the related work in the next section. Then, we systematically outline the technical framework of our approach in Section III. Subsequently, we introduce the Enhanced Low-light Night Vision Dataset (ELND) in Section IV. Furthermore, Section V presents the experimental procedures and provides a comprehensive analysis of the results. Finally, we summarize the whole paper in Section VI.

II. RELATED WORK

This section reviews supervised deep learning approaches for low-light image enhancement, as well as relevant techniques based on the Fast Fourier Transform (FFT).

A. Supervised deep learning-based low-light enhancement

Current supervised deep learning methods for low-light enhancement predominantly build upon convolutional neural networks (CNNs). Early CNN-based approaches [15]–[17] relied on pixel-wise hierarchical frameworks to adjust illumination. Yet, they often suffered from noise accumulation as network depth increased, due to the difficulty in separating signal from noise. To alleviate noise amplification while preserving details, subsequent works introduced several refinements: frequency-aware decomposition modules [16], gradient-guided feature learning [18], [19], and multi-stage or multi-branch architectures [20]–[24]. These adaptations enable more robust

enhancement across varying light conditions. Nonetheless, a fundamental limitation persists: CNNs are inherently limited in modeling long-range dependencies across spatial and channel dimensions, which is crucial for global illumination adjustment and structural recovery.

Inspired by the success of Transformers in high-level vision tasks [9], [25], [26], researchers have also adapted them to low-light image enhancement [23], [26]–[31]. For instance, Xu et al. [26] introduced an SNR-aware prior to adaptively process pixels by fusing long- and short-range features, mitigating noise propagation. Other works, such as MPC-NET [27], have also demonstrated the effectiveness of Transformer-based designs. However, the quadratic complexity of standard self-attention limits their scalability to high-resolution images. To improve efficiency, several variants have been proposed, including improved multi-head attention and feedforward modules [28], axis-based attention with gating mechanisms [29], and window-based attention in Uformer [30] and SwinIR [31]. While these strategies reduce computation, they largely retain the core multi-head self-attention (MHSA) operation, which is known to contain feature redundancy [30]–[35] and may constrain representation capacity. In contrast, the MFCA component proposed in this paper is designed to explicitly reduce both computational cost and feature redundancy, offering a more efficient and representative attention mechanism.

B. FFT-based low-light enhancement

The Fast Fourier Transform (FFT) enables a frequency-domain representation of images, offering a complementary perspective to spatial modeling and providing a global receptive field. This has led to its growing adoption in deep learning for vision tasks [36]–[39]. In the context of low-light enhancement, frequency-domain approaches often decompose an image into amplitude and phase components. Prior work has established that the amplitude component correlates strongly with image lightness and exposure [37], [40], while noise characteristics can also be analyzed in Fourier space [41]. Consequently, several methods [41]–[43] perform enhancement by primarily manipulating amplitude to adjust global illumination and capture long-range dependencies. However, their reliance on direct amplitude manipulation provides limited modeling of how information and noise are distributed across different frequency bands. In this work, we propose a multi-band frequency filtering mechanism designed for extreme low-light night vision images. Instead of directly manipulating the amplitude spectrum, our module uses multi-scale band-pass filters to decompose frequency-domain features into multiple bands and then adaptively fuses the resulting band-wise features. This strategy enables more comprehensive frequency-domain modeling, strengthens global context modeling, and better preserves structural information.

III. METHODOLOGY

In this section, we present our piecewise iterative enhancement framework and the proposed Multi-band Iterative Enhancement Former (MIEFormer) with its core MFCA module.

The framework is motivated by an analysis of the frequency-domain characteristics of extremely low-light images and their signal-to-noise ratio (SNR) under varying illumination conditions. Guided by this framework, we detail the overall architectures of MIEFormer and MFCA (Fig. 2). Finally, we define the loss function used for supervised training.

A. Preliminary: Frequency-Domain Analysis

We analyze low-light images under varying illumination levels using the 2D discrete Fourier transform (2D-DFT). Given an input image x of size $H \times W$, its DFT is defined as Eq. (1):

$$X(u, v) = \frac{1}{\sqrt{HW}} \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} \chi(h, w) e^{-j2\pi(\frac{h}{H}u + \frac{w}{W}v)}, \quad (1)$$

where h and w denote the coordinates in spatial domain, while u, v represent the coordinates in frequency domain. The symbol j is the imaginary unit. $X(u, v)$ is a complex-valued function, whose amplitude and phase components are given as follows:

$$\begin{aligned} A(X(u, v)) &= \sqrt{R^2(X(u, v)) + I^2(X(u, v))}, \\ P(X(u, v)) &= \arctan \frac{I(X(u, v))}{R(X(u, v))}, \end{aligned} \quad (2)$$

where $A(\cdot)$ and $P(\cdot)$ represent the amplitude and phase components, respectively, while $R(\cdot)$ and $I(\cdot)$ denote the real and imaginary parts of $X(u, v)$.

Based on the above 2D-DFT, we examine images from our ELND dataset across 10^{-1} to 3×10^{-4} lux (see Fig. 1) and obtain three key observations that directly motivate our piecewise iterative design: (a) As illumination decreases, the image SNR declines correspondingly, with the extent of degradation differing across illumination stages. (b) In the frequency domain, lower illumination leads to noticeable distortion and information loss in the amplitude spectrum. (c) When enhancing extremely low-light images, high-frequency information from a single frequency band is often too sparse for complete detail recovery, whereas integrating multi-band high-frequency components results in significantly better reconstruction.

Given these findings, directly learning a mapping from extremely low-light to normal-light images is highly challenging. To address this, we propose a piecewise successive iterative residual learning framework. Specifically, we first discretize the continuous illumination spectrum from extremely low light to normal light into multiple adjacent intervals. Within each interval, a dedicated residual learning module captures illumination-specific characteristics and incremental information. Enhancement is then achieved through iterative refinement across these intervals, progressively improving the overall image quality. Based on the residual learning approach proposed in [44], the low-light image enhancement process is modeled as a residual learning task and can be formulated as follows:

$$Y_{\text{norm}} = I + P(I) - n, \quad (3)$$

where $Y_{\text{norm}} \in \mathbb{R}^{W \times H \times 3}$ denotes the normal-light (NL) image, $I \in \mathbb{R}^{W \times H \times 3}$ represents the low-light (LL) input image, $P(\cdot)$

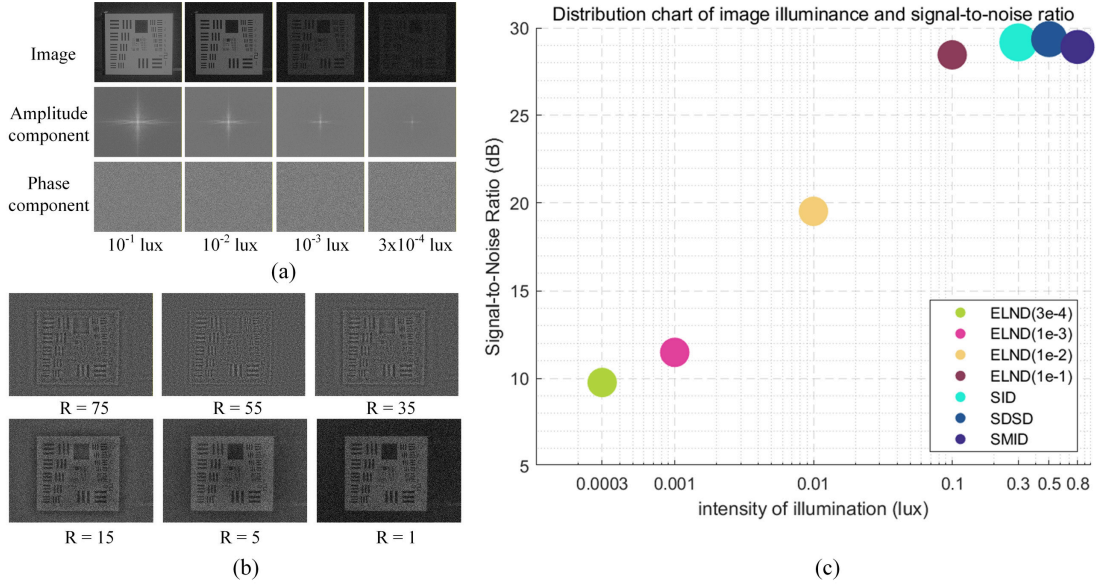


Fig. 1. Observation of low-light night vision images: (a) Original images and their Fourier transform components at various illumination levels; (b) High-frequency information of the ideal high-pass filter with different radii R during enhancement at 1×10^{-3} lux; (c) Distribution of illumination intensity and noise signal-to-noise ratio in a low-light image dataset.

signifies the enhancing operator that estimates the residual between the NL and LL images, and $n \in \mathbb{R}^{W \times H \times 3}$ represents the noise to be removed. The N adjacent intervals can be represented as $\{(Y_{\text{norm}}, Y_{\text{lux}}^{N-1}), (Y_{\text{lux}}^{N-1}, Y_{\text{lux}}^{N-2}), \dots, (Y_{\text{lux}}^1, I)\}$. The corresponding residual learning for each illuminance interval can be expressed as:

$$\begin{aligned} Y_{\text{lux}}^1 &= I + P_1(I) - n_1, \\ Y_{\text{lux}}^2 &= Y_{\text{lux}}^1 + P_2(Y_{\text{lux}}^1) - n_2, \\ &\vdots \\ Y_{\text{norm}} &= Y_{\text{lux}}^{N-1} + P_N(Y_{\text{lux}}^{N-1}) - n_N. \end{aligned} \quad (4)$$

Substituting Eq. (4) into Eq. (3) yields:

$$\begin{aligned} Y_{\text{norm}} - I &= (Y_{\text{norm}} - Y_{\text{lux}}^{N-1}) \\ &\quad + (Y_{\text{lux}}^{N-1} - Y_{\text{lux}}^{N-2}) + \dots + (Y_{\text{lux}}^1 - I) \\ &= \sum_{i=2}^N (P_i(Y_{\text{lux}}^{i-1}) - n_i) + P_1(I) - n_1. \end{aligned} \quad (5)$$

Defining $Y_{\text{lux}}^0 = I$ and $T_i(Y_{\text{lux}}^{i-1}, n_i) = P_i(Y_{\text{lux}}^{i-1}) - n_i$, the above expression can be simplified to $Y_{\text{norm}} - I = \sum_{i=1}^N T_i(Y_{\text{lux}}^{i-1}, n_i)$, where $T_1(Y_{\text{lux}}^0, n_1) = P_1(I) - n_1$. Low-light enhancement can therefore be formulated as learning operators $T_i(\cdot)$ that balance illumination enhancement and noise suppression.

In practical applications, as the value of N increases, the enhancement quality improves, but substantial artifacts, blurring, and noise are simultaneously introduced. This phenomenon arises because the operator $T_i(\cdot)$, while effectively achieving enhancement and noise suppression, inherently introduces bias. As N increases, such bias accumulates gradually. Furthermore, without proper constraints, the accumulated error is further amplified, leading to significant degradation

and inconsistency in the enhancement performance. To address this issue, we propose a piecewise learning paradigm. Specifically, we modify the ground-truth (GT) image and extend the single GT reference into a sequence of images captured under diverse illumination conditions. $Y_{GT} = Y_{\text{norm}}, \bar{Y}_{\text{lux}}^n, \bar{Y}_{\text{lux}}^{n-1}, \dots, \bar{Y}_{\text{lux}}^1, \bar{Y}_{\text{lux}}^i \in \mathbb{R}^{W \times H \times 3}$. The optimization of the enhancement operator $T_i(\cdot)$ corresponding to each illumination interval is formulated as Eq. (6).

$$T_i = \arg \min_{T_i} \left(\|\bar{Y}_{\text{lux}}^i - (Y_{\text{lux}}^{i-1} + T_i(Y_{\text{lux}}^{i-1}, n_i))\|_2 + \Omega(T_i) \right). \quad (6)$$

Here, $\|\cdot\|_2$ represents the L2 norm, and $\Omega(T_i)$ denotes the regularization term. By imposing an additional constraint upon each illumination region, this scheme further mitigates modeling bias and suppresses its cumulative drift, thereby stabilizing the overall enhancement performance consistently. The design of the operator $T_i(\cdot)$ takes both enhancement and noise suppression into account, as detailed in Eq. (7):

$$T_i(Y_{i-1}, n_i) = \delta_i(Y_{i-1}) + \sigma_i(\delta_i(Y_{i-1}), n_i), \quad (7)$$

where $\delta_i(\cdot)$ represents the residual enhancement function between images captured under varying illumination conditions, whereas $\sigma_i(\cdot)$ signifies the function employed for noise suppression.

B. Overall Architecture of MIEFormer

Based on the above motivation, we propose our MIEFormer, as shown in Fig. 2. The model follows a piecewise iterative enhancement pipeline. At each stage, an Enhancement Segment Module (ESM) performs supervised enhancement within a specific illumination interval. Each ESM is designed to not only adaptively improve image quality under

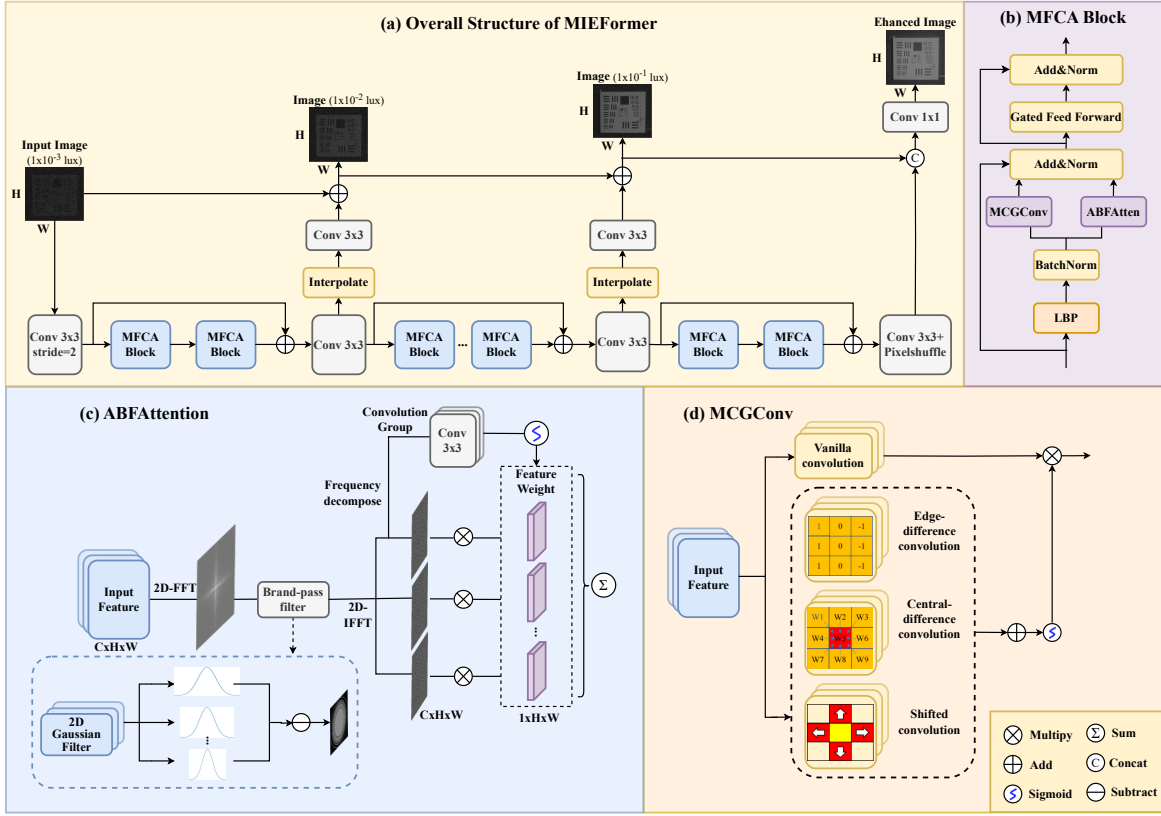


Fig. 2. (a) Overall Structure of MIEFormer. The network includes multiple Enhanced Segment Modules (ESMs), each with several MFCA blocks; (b) Structure of MFCA Block; (c) Structure of ABFAAttention; (d) Structure of MCGConv.

its target illumination but also suppress noise introduced during processing, ensuring robustness across varying light conditions. To maintain a simple yet effective architecture, each ESM stacks multiple MFCA blocks, with the number of blocks determined according to the enhancement difficulty at the corresponding stage. Each MFCA block follows a Transformer-style architecture and offers three main advantages: (1) multi-band frequency-domain fusion replaces multi-head self-attention, reducing computational overhead while retaining global-context modeling; (2) piecewise residual learning divides the difficult enhancement task into manageable stage-specific subproblems that jointly balance illumination enhancement and noise suppression; and (3) enhanced extraction of local edges and gradients improves the discrimination of informative details from noise.

Taking an image with illuminance of 3×10^{-4} lux as an example, we denote the input image as $X \in \mathbb{R}^{W \times H \times 3}$, where H and W represent its height and width, respectively. First, shallow features F_0 are extracted via a 3×3 convolution with stride 2, followed by leaky ReLU activation. The specific calculation of the convolution operation is given by Eq. (8):

$$F_0 = \text{LeakyReLU} \left(\text{conv}^{3 \times 3, \text{stride}=2}(X) \right), \quad F_0 \in \mathbb{R}^{\frac{W}{2} \times \frac{H}{2} \times 32}. \quad (8)$$

The shallow feature is then fed into the core network, which integrates multiple ESMs. We detail an ESM designed for the illumination range of 3×10^{-4} to 1×10^{-1} lux. Each ESM comprises two sequential components: feature enhancement and feature recovery. The feature enhancement component

employs two MFCA blocks followed by a 3×3 convolution, with the MFCA output added to the convolutional output via a residual connection. This design focuses on illumination-specific enhancement and noise suppression. Subsequently, the feature recovery component upsamples the refined features via interpolation and applies a final 3×3 convolution to generate the enhanced ESM, the output corresponds to the enhanced result under 1×10^{-3} lux illumination. Following the same piecewise enhancement principle, subsequent ESMs progressively produce results at 1×10^{-2} lux and finally at 1×10^{-1} lux. The overall procedure is formalized in Eq. (9):

$$\begin{aligned} F_i &= \text{conv}^{3 \times 3}(\text{MFCA} \cdots (\text{MFCA}(F_{i-1})) + F_{i-1}), \\ P_i &= \text{conv}^{3 \times 3}(\text{Interpolate}(F_{i-1})) + P_{i-1}, \\ i &= 1, \dots, N, \quad P_0 = X, \end{aligned} \quad (9)$$

where F_i represents the enhanced feature map within the i -th ESM, P_i denotes the predicted image under specific illuminance conditions, and N signifies the total number of ESMs.

Finally, the enhanced feature map under 1×10^{-1} lux illumination is passed through a final, refined ESM. Diverging from previous stages, this module prioritizes high-fidelity feature recovery, drawing inspiration from super-resolution techniques. Specifically, we replace the previous upsampling interpolation and 3×3 convolution with a sub-pixel convolution layer (pixelshuffle) followed by a 1×1 convolutional layer. This revised structure not only recovers resolution but

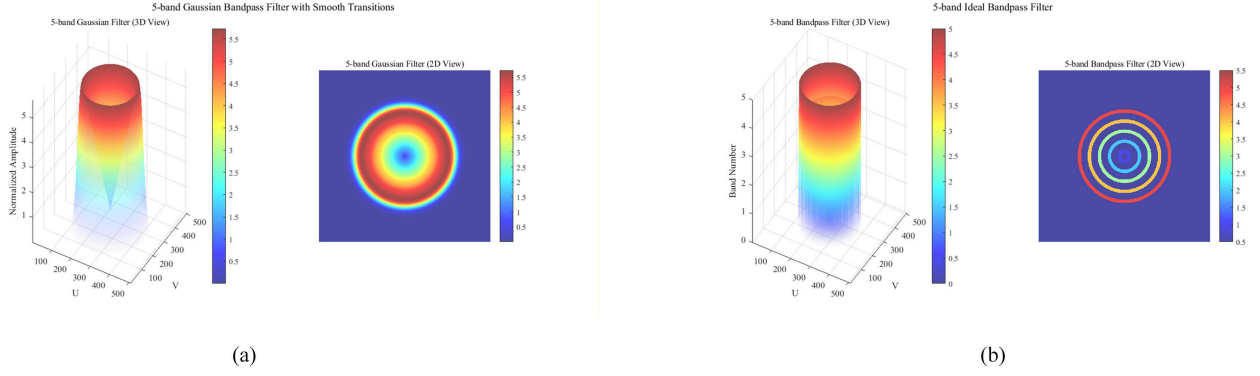


Fig. 3. Multi-scale Gaussian and Ideal Bandpass Filters. (a) 5-order Gaussian Bandpass Filter (b) 5-order ideal bandpass filter.

also enhances fine-grained details, thereby mitigating artifacts and yielding superior final visual quality.

$$\begin{aligned} F_{gt} &= \text{conv}^{3 \times 3}(\text{MFCA} \cdots (\text{MFCA}(F_N)) + F_N), \\ P_{gt} &= \text{conv}^{1 \times 1}(\text{PixelShuffle}(F_{gt})) + P_N, \end{aligned} \quad (10)$$

where F_N signifies the predicted feature map under the 1×10^{-1} lux illuminance condition. Correspondingly, P_{gt} and P_N indicate the final enhanced image and the predicted enhanced image at the 1×10^{-1} lux illuminance.

C. Multi-frequency and Content-guided Attention Block

Building upon the overall architecture, this section further details the core component of our MIEFormer, i.e., the MFCA block. As derived from Eq. (5) and depicted in Fig. 2(b), the MFCA block is designed to perform two critical functions: learning residual enhancements specific to an illumination interval and suppressing noise introduced during the enhancement process.

As shown in Fig.2(b), the MFCA block adopts a structure analogous to a Transformer block. First, a Lighting Back Projection (LBP) module [44] refines illumination-related features spatially. These features are then projected into the frequency domain via FFT. The subsequent Adaptive Band-pass Filtering Attention (ABFAttention) module harnesses a multi-scale frequency decomposer to split the spectrum into band-wise components. Incorporating frequency-domain prior knowledge, it adaptively weights and fuses these bands to strengthen long-range dependency modeling and refine global illumination control.

In parallel, a Multi-branch Content-guided Convolution (MCGConv) module enhances local feature discrimination, particularly between edge structures and noise patterns, thereby improving noise suppression. Finally, a gated feed-forward network is applied to promote parameter sharing, increase nonlinear representational capacity, and improve the overall robustness of the block.

(1) Adaptive Band-pass Filtering Attention

The FFT-based modules in [41]–[43] enhance brightness and restore color fidelity primarily by amplifying amplitude-spectrum features. Under extremely low-light conditions, however, direct amplitude amplification may also amplify noise

and introduce spurious artifacts. Although ABFAttention likewise uses FFT operations, it serves a different purpose: it decomposes features into multiple frequency bands and adaptively fuses them to refine useful information while suppressing band-specific noise. This strategy improves robustness across diverse illumination conditions. As depicted in Fig. 2(c), the ABFAttention module employs a Multi-Scale Frequency Feature Decomposer to partition input features into distinct frequency bands, then performs adaptive, Multi-band Frequency-aware Fusion.

Multi-Scale Frequency Feature Decomposer. Band-pass filters are widely used in frequency-domain analysis to isolate signal components within specific frequency ranges. Compared to high- and low-pass filters, they allow for more precise frequency interval selection. Given this property, we design a frequency feature decomposer based on a multi-scale Gaussian band-pass filter, which offers smoother frequency transitions than an ideal (rectangular) band-pass filter, as shown in Fig. 3. In practice, the input feature is first transformed into the frequency domain via 2D-FFT. The multi-scale Gaussian band-pass filter is then applied in the frequency domain to extract components across different frequency bands. Finally, the 2D Inverse Fourier transform (2D-IFFT) is used to convert each filtered component back to the spatial domain, resulting in a set of features decomposed across multiple frequency bands.

More formally, let $f \in \mathbb{R}^{w \times h \times c}$ denote the input feature map, where w , h , and c represent the width, height, and number of channels of the feature map, respectively. $F_{2D}(\cdot)$ and $F_{2D}^{-1}(\cdot)$ denote the 2D Fourier transform and its inverse transform, respectively. σ_{lf} and σ_{hf} denote the standard deviations of the Gaussian functions corresponding to the start frequency and cutoff frequency, respectively, with $\sigma_{hf} > \sigma_{lf}$, and (u, v) being the frequency-domain coordinates. The entire process is given by Eq. (11).

$$\begin{aligned} H_{lf}(u, v; \sigma_{lf}) &= \exp\left(-\frac{u^2 + v^2}{2\sigma_{lf}^2}\right), \\ H_{hf}(u, v; \sigma_{hf}) &= \exp\left(-\frac{u^2 + v^2}{2\sigma_{hf}^2}\right), \\ H_{bp}(u, v; \sigma_{lf}, \sigma_{hf}) &= \exp\left(-\frac{u^2 + v^2}{2\sigma_{hf}^2}\right) - \exp\left(-\frac{u^2 + v^2}{2\sigma_{lf}^2}\right), \\ \{f_{bp}^1, f_{bp}^2, \dots, f_{bp}^n\} &= F_{2D}^{-1}(F_{2D}(f) \times H_{bp}(u, v)), \end{aligned} \quad (11)$$

where H_{lf} , H_{hf} and H_{bp} denote the 2D Gaussian Low-Pass Filter (2D GLPF) for the start frequency, the 2D GLPF for the

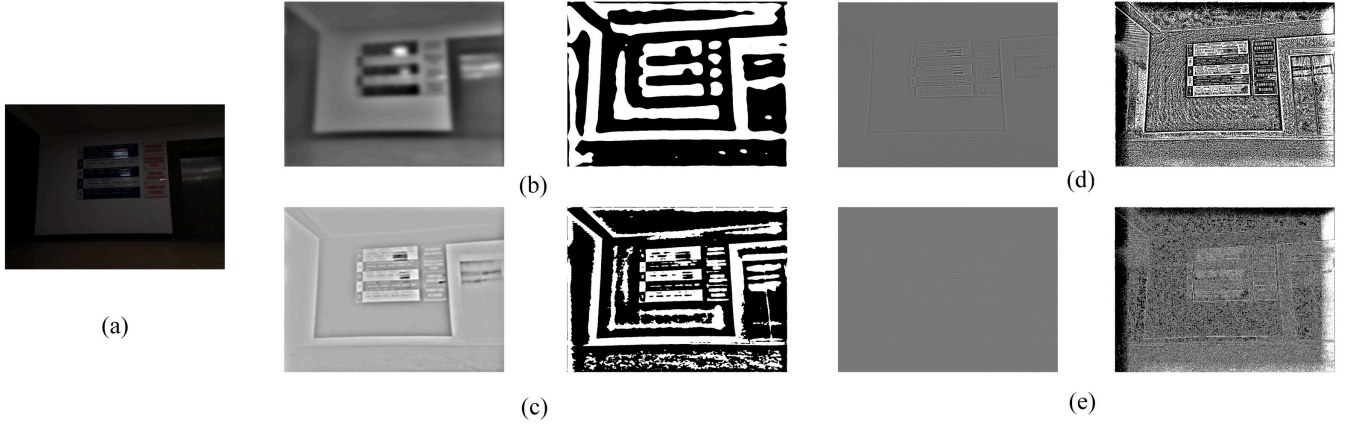


Fig. 4. Images of different sub-frequency bands, along with their corresponding prior map. (a) Input Image; (b) Low-frequency band image and its prior map; (c) low-to-mid frequency band images and their prior map; (d) Mid-to-high frequency band images and their prior map; (e) High-frequency band image and its prior map.

cutoff frequency, and the 2D Gaussian Band-Pass Filter (2D GBPF), respectively. $\{f_{bp}^1, f_{bp}^2, \dots, f_{bp}^n\}$ represent the series of feature sets corresponding to different frequency bands obtained after decomposition. We divide the frequency domain into four bands whose start and cutoff frequencies are (3, 7), (7, 13), (13, 19), and (19, 21), respectively.

Multi-band Frequency-aware Fusion. We treat multi-scale band-pass filters as frequency-band samplers, where different start/cutoff frequencies and bandwidths extract information from distinct frequency bands. As the sampling frequency increases, the captured information shifts from dense low-frequency components to sparse high-frequency details. A learnable basis function is introduced to generate frequency-specific prior maps for each band, as illustrated in Fig. 4, with the process described as follows:

Let $f \in \mathbb{R}^{w \times h \times c}$ denote the input feature map, where w , h , and c represent the width, height, and number of channels of the feature map, respectively. $H_{bp}(\cdot)$ denotes the frequency band decomposer constructed in the previous section. To integrate the decomposition operation and the learnable basis function into a unified framework, the specific implementation of generating frequency-specific prior maps is given by Eq. (12).

$$\begin{aligned} \{f_{bp}^1, f_{bp}^2, \dots, f_{bp}^n\} &= H_{bp}(f), \\ m_{bp}^i &= \text{sigmoid}\left(\text{conv}^{3 \times 3}(\text{conv}^{3 \times 3}(f_{bp}^{i+1}))\right), \\ &\quad i = 1, \dots, n-1, \end{aligned} \quad (12)$$

where m_{bp}^i represents the frequency prior map corresponding to the frequency band f_{bp}^i , and let n denote the number of frequency bands.

As shown in Fig. 4, the generated prior maps exhibit distinct characteristics across different frequency bands. Specifically, the overall brightness representation transitions from coarse to fine scales, while structural details evolve from being imperceptible in dark areas, to becoming visible across the entire image, and finally to being dominated by amplified noise in the highest bands. This variation can be formulated as a weighted fusion of the different frequency-band components.

By adaptively adjusting these band-wise weights, our model achieves precise brightness adjustment and targeted noise suppression within specific frequency ranges, as formulated in Eq. (13).

$$f_{\text{use}} = f_{bp}^1 + \sum_{i=1}^{n-1} (1 - m_{bp}^i) f_{bp}^{i+1}, \quad (13)$$

where f_{use} denotes the fused feature map.

(2) Multi-branch Content-guided convolution

Although the ABFAAttention module effectively captures long-range dependencies, repeated enhancement may progressively reduce the distinction between meaningful details and amplified noise. This reduced discriminability can introduce artifacts into the restored image, such as the loss of fine details and blurred edges.

Inspired by edge-aware prior methods for detail restoration [41], we propose a Multi-branch Content-guided Convolution (MCGConv) module, depicted in Fig. 2(d). MCGConv improves the discrimination of informative details from noise by exploiting the complementary properties of standard and differential convolutions during local feature extraction. Standard convolution aggregates pixel values within a local receptive field and is effective at capturing contextual appearance, but this aggregation may weaken object boundaries and subtle details. By contrast, differential convolution responds strongly to local intensity differences and is therefore well suited to capturing edges, textures, and other fine structural cues. This module adopts a multi-branch architecture: multiple parallel differential convolutions first generate content-guided prior maps, which subsequently serve as soft masks to modulate the outputs of standard convolutional branches. This design reinforces the separation of structural details from noise.

Differential Convolution (DC) enhances convolutional representation by integrating gradient-like local descriptors [45], [46]. While variants such as Central Difference Convolution (CDC) [45], [47] and Directional Difference Convolution (DDC) [46], [47] are adept at capturing fine-grained features, their ability to handle rotation and to fuse multi-scale information remains constrained. To address these limitations, we

Algorithm 1 Adaptive Band-Pass Filtering Attention (ABFAt-tention)

Require: Input feature map $x \in \mathbb{R}^{B \times C \times H \times W}$; attention feature x_{attn} (default: x); band frequency parameters $K = [k_1, \dots, k_N]$; spatial group G ; kernel size k_{sp}

Ensure: Fused multi-band feature y

- 1: Initialize learnable spatial convolutions $\{\text{Conv}_i\}_{i=0}^N$
- 2: Initialize $\text{Conv}_i.w = 0, \text{Conv}_i.b = 0$
- 3: **Forward**(x, x_{attn})
- 4: **if** x_{attn} is None **then**
- 5: $x_{\text{attn}} \leftarrow x$
- 6: **end if**
- 7: $x_{\text{list}} \leftarrow []$
- 8: $x_{\text{pre}} \leftarrow x$
- 9: **Step 1: Global FFT transform**
- 10: $X_{\text{fft}} \leftarrow \text{FFTSHIFT}(\text{FFT2}(x, \text{norm} = \text{'ortho'}))$
- 11: **Step 2: Recursive band-pass decomposition**
- 12: **for** $j = 1, 2, \dots, N$ **do**
- 13: *// Construct low-frequency Gaussian filter*
- 14: $\text{Filter}_l \leftarrow \text{Gaussian}(k_{j-1})$
- 15: *// Extract low-frequency component*
- 16: $x_{\text{low}} \leftarrow \text{IFFT2}(\text{IFFTSHIFT}(X_{\text{fft}} \odot \text{Filter}_l), \text{norm} = \text{'ortho'}).real$
- 17: *// Compute high-frequency residual*
- 18: $x_{\text{high}} \leftarrow x_{\text{pre}} - x_{\text{low}}$
- 19: $x_{\text{pre}} \leftarrow x_{\text{low}}$
- 20: *// Generate spatial adaptive gating weights*
- 21: $\alpha_j \leftarrow 2\sigma(\text{Conv}_{j-1}(x_{\text{attn}}))$
- 22: *// Group-wise feature modulation*
- 23: $x_{\text{group}}^{(j)} \leftarrow \alpha_j \odot \text{trans}_G(x_{\text{high}}^{(j)})$
- 24: $x_{\text{list}} \leftarrow x_{\text{list}} \cup \{\text{trans}_C(x_{\text{group}}^{(j)})\}$
- 25: **end for**
- 26: **Step 3: Modulate final low-frequency component**
- 27: $\alpha_{N+1} \leftarrow 2\sigma(\text{Conv}_N(x_{\text{attn}}))$
- 28: $x_{\text{group}}^{(N+1)} \leftarrow \alpha_{N+1} \odot \text{trans}_G(x_{\text{pre}}^{(N+1)})$
- 29: $x_{\text{list}} \leftarrow x_{\text{list}} \cup \{\text{trans}_C(x_{\text{group}}^{(N+1)})\}$
- 30: **Step 4: Aggregate all frequency bands**
- 31: $y \leftarrow \sum_{j=1}^{N+1} x_{\text{list}}[j]$
- 32: **return** y

propose a Shifted-Central Difference Convolution (SCDC). As illustrated in Fig. 5, SCDC first shifts the feature map along four spatial directions (up, down, left, right) to emulate non-central receptive patterns. It then computes differential correlations within each shifted local window, thereby enhancing rotation awareness and multi-scale fusion.

Implementation of MCGConv. By applying CDC, DDC, and SCDC operations to the input features, the model extracts complementary fine-grained representations. These features are then aggregated and passed through a sigmoid function to produce a spatially adaptive mask map, which fuses local descriptive and gradient information. Acting as a content-aware prior, this mask is multiplied element-wise with the output of a standard convolutional branch, thereby improving the preservation and sharpness of structural details. The overall operation is formulated in Eq. (14).

$$f' = \text{conv}^{3 \times 3}(f) \times \text{sigmoid}(\text{CDC}^{3 \times 3}(f) + \text{DDC}^{3 \times 3}(f) + \text{SCDC}^{3 \times 3}(f)). \quad (14)$$

Algorithm 2 Shifted-Central Difference Convolution (SCDC)

Require: Input feature map $x \in \mathbb{R}^{B \times C \times H \times W}$; learnable convolution weight $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times 3 \times 3}$; bias b ; stride s ; padding p

Ensure: Output feature map $y \in \mathbb{R}^{B \times C_{\text{out}} \times H \times W}$

- 1: $y_0 \leftarrow \text{Conv2D}(x, W, \text{bias} = b, \text{stride} = s, \text{padding} = p)$
// Standard convolution output
- 2: $\xi \leftarrow \{1, 3, 5, 7\}$ *// Shifted center indices corresponding to w_2, w_4, w_6, w_8*
- 3: **for** $i = 1, 2, \dots, |\xi|$ **do**
- 4: $c \leftarrow \xi[i]$
- 5: $W_{\text{flat}} \leftarrow \text{reshape}(W, [C_{\text{out}}, C_{\text{in}}, 9])$
- 6: $w_c \leftarrow W_{\text{flat}}[:, :, c]$ *// Extract shifted-center weight*
- 7: $W_{\text{diff}} \leftarrow W_{\text{flat}} - w_c$ *// Central difference via broadcast*
- 8: $W_{\text{diff}}[:, :, c] \leftarrow 0$ *// Zero out shifted center*
- 9: $W_{\text{diff}} \leftarrow \text{reshape}(W_{\text{diff}}, [C_{\text{out}}, C_{\text{in}}, 3, 3])$
- 10: $y_i \leftarrow \text{Conv2D}(x, W_{\text{diff}}, \text{bias} = \text{None}, \text{stride} = s, \text{padding} = p)$
- 11: **end for**
- 12: $y \leftarrow y_0$
- 13: **for** $i = 1, 2, \dots, |\xi|$ **do**
- 14: $y \leftarrow y + y_i$ *// Aggregate shifted difference branches*
- 15: **end for**
- 16: **return** y

Here, $f \in \mathbb{R}^{w \times h \times c}$ denotes the input feature map, where w , h , and c represent the width, height, and number of channels of the feature map, respectively. $\text{CDC}(\cdot)$, $\text{DDC}(\cdot)$, and $\text{SCDC}(\cdot)$ denote the CDC, DDC, and SCDC operations, respectively. f' denotes the enhanced feature map.

D. Loss Function

Building on its iterative, piecewise enhancement framework, MIEFormer incorporates a dedicated, supervised Enhancement Segment Module (ESM) at each stage. Consequently, the total training loss function is formulated as a weighted summation of the individual losses from all stages.

The overall loss consists of two terms: an illumination loss and a reconstruction loss. The illumination loss guides the piecewise iterative adjustment of lighting conditions, while the reconstruction loss ensures the detail fidelity and perceptual quality of the final output.

The illumination enhancement is realized through a supervised, coarse-to-fine process via piecewise iterative residual learning. At each stage, we employ a combined loss based on Structural Similarity (SSIM) and Mean Absolute Error (MAE). To precisely define this hybrid loss and ensure it effectively guides the residual learning of each stage, we formulate it as follows in Eq. (15):

$$\begin{aligned} L_{\text{illu}}^i &= L_{\text{ssim}} + L_{\text{mae}}, \\ L_{\text{ssim}} &= 1 - \text{SSIM}(I_{\text{gt}}, I_{\text{pred}}), \\ L_{\text{mae}} &= \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |I_{\text{gt}}(i, j) - I_{\text{pred}}(i, j)|, \end{aligned} \quad (15)$$

where L_{illu}^i denotes the i -th illuminance loss function, L_{ssim} denotes the SSIM loss function, L_{mae} denotes the MAE loss

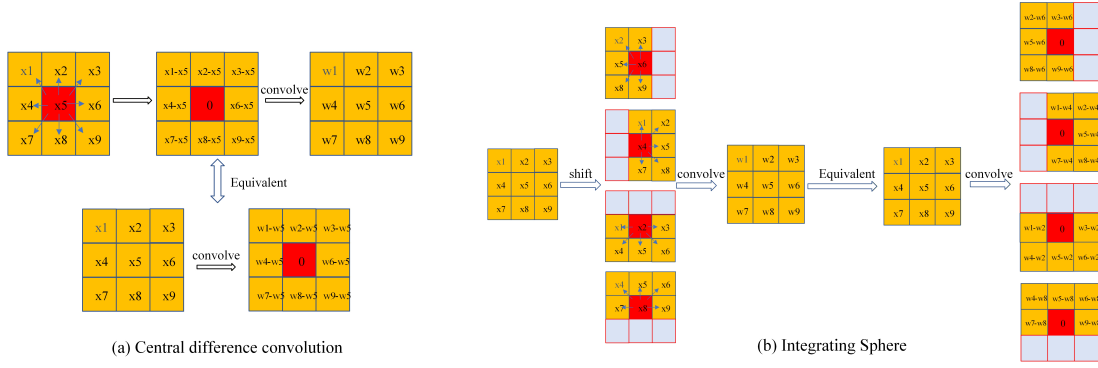


Fig. 5. Central difference convolution and shifted-Central difference convolution (a) Central difference convolution; (b) shifted-Central difference convolution

function, I_{gt} and I_{pred} denote the ground truth image and predicted image, respectively, and $SSIM(\cdot)$ represents the structural similarity function.

The reconstruction loss is designed to ensure that the illumination-enhanced output closely matches the reference image, focusing on detail fidelity and perceptual quality. It combines a perceptual loss with the Mean Squared Error (MSE) loss, and its complete formulation is provided in Eq. (16).

$$L_{recon} = L_{lips} + L_{mse},$$

$$L_{lips} = \sum_{l=1}^L \frac{1}{H \times W} \|\mathbf{w}_l \odot (\varphi_l(I_{gt}) - \varphi_l(I_{pred}))\|_2^2, \quad (16)$$

$$L_{mse} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (I_{gt}(i, j) - I_{pred}(i, j))^2,$$

where L_{recon} represents the reconstruction loss function, L_{lips} denotes the perceptual loss function, L_{mse} denotes the mean squared error (MSE) loss function, $\varphi_l(\cdot)$ denotes the feature extraction function of the layer in the pre-trained VGG19 network, $\mathbf{w}_l$ is the weight corresponding to the layer, $\odot$ indicates element-wise multiplication.

Finally, the overall loss function is expressed as Eq. (17):

$$L = \lambda \sum_{i=1}^n L_{illu}^i + \beta L_{recon}, \quad (17)$$

where $\lambda = 0.6$, $\beta = 0.4$.

IV. ENHANCED LOW-LIGHT NIGHT VISION DATASET

Imaging under extremely low-light conditions presents key challenges, including complex noise distributions, non-uniform illumination, and varied scene content. Existing public datasets are predominantly limited to illuminance levels above 10^{-2} lux, seldom covering the extreme low-light regime below 10^{-3} lux.

Moreover, synthetically generated noise often fails to accurately replicate real sensor noise patterns, while available scene content rarely reflects typical night-vision environments.

To address these limitations, we introduce the Enhanced Low-light Night Vision Dataset (ELND), a dedicated benchmark for extreme low-light image enhancement. The ELND consists of two distinct subsets:

- 1) Indoor scenes: Images are captured in a controlled optical darkroom at five precisely calibrated illuminance levels: 10^{-1} , 10^{-2} , 3×10^{-3} , 10^{-3} , and 3×10^{-4} lux. The target board has a black background and contains white uppercase English letters, white-filled regular polygons (triangles, rectangles, diamonds, and regular pentagons), and USAF-1951 and ISO resolution charts. The foreground targets are captured both individually and in composite arrangements.
- 2) Outdoor scenes: Images are collected under three natural low-light conditions: moonlight, starlight, and overcast conditions without direct illumination. The moonlight and starlight scenes contain the sky and mountain peaks, whereas the overcast scenes contain pedestrians and vehicles. The acquisition location is changed between sessions to increase scene diversity.

ELND comprises 29 indoor image sequences, totaling 9,135 images. Each sequence contains 63×5 images (63 images at each of five illuminance levels). The outdoor subset comprises 1,200 images: 620 moonlight images, 360 starlight images, and 220 images captured without direct illumination. Figs. 6 and 7 present the data distribution and representative samples of ELND, respectively.

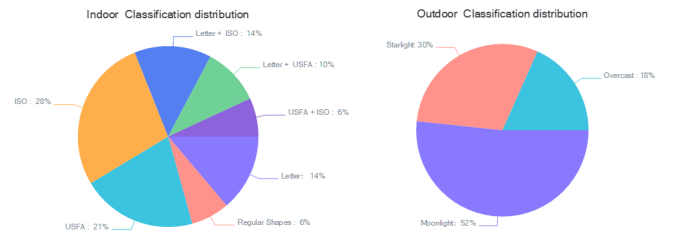


Fig. 6. Data Distribution for ELND

Compared to existing low-light datasets, ELND offers more controlled acquisition and reliable annotation. In terms of data capture, imaging under extremely low illumination is inherently limited by hardware sensitivity and ambient lighting. Commonly used cameras such as the Sony A7RIII or Canon EOS R can theoretically operate down to around 1×10^{-3} lux by adjusting aperture and shutter speed, yet in practice rarely reach such low illuminance reliably under real-world constraints.

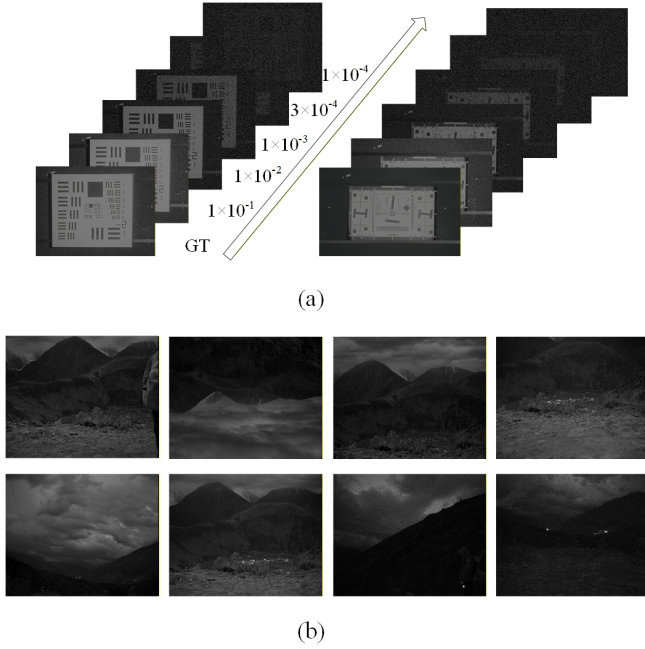
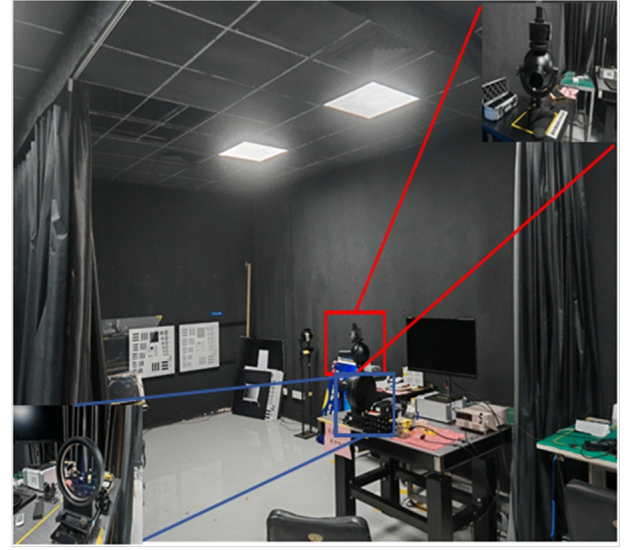


Fig. 7. Examples from the ELND dataset. (a) Indoor Environments, image sequences were captured at various illumination levels: 1×10^{-1} lux, 1×10^{-2} lux, 1×10^{-3} lux, 3×10^{-3} lux, and 3×10^{-4} lux. (b) Outdoor Environments, capturing images outdoors under moonlight, starlight, and in complete darkness.

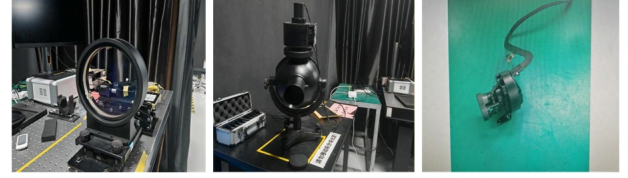
Data Acquisition. Lighting conditions differ substantially between indoor and outdoor environments in terms of uniformity, stability, and spatial coverage. We therefore designed and implemented two separate acquisition protocols.

- 1) **Indoor Acquisition Protocol.** The controlled indoor environment provides stable lighting with minimal external interference, enabling precise calibration for quantitative evaluation. As illustrated in Fig. 8, we built a dedicated optical platform comprising (1) an integrating sphere for stable and adjustable illumination; (2) a collimating objective lens with a focal length of $f = 4.25$ m, an effective aperture of $D_0 = \Phi 200$ mm, and a field of view of $2\omega \geq 12^\circ$; and (3) a custom night-vision camera equipped with a Brigades sensor. The camera has an output resolution of 1280×1024 pixels, a depth of field from 2 m to infinity, image distortion below 3%, horizontal and vertical fields of view of 40.35° and 32.14° , respectively, and an imaging sensitivity range of 1×10^{-4} –10 lux. Using this setup, we captured graded image sequences at five illumination levels.
- 2) **Outdoor Acquisition Protocol.** Outdoor settings present expansive scenes with complex, unstable lighting and diverse interference factors—conditions that better reflect real-world application challenges. Using the same low-light night-vision camera, we mounted it on a fixed tripod and collected footage under three natural illumination conditions: moonlight, starlight, and overcast (no direct light).

Ground-Truth Generation. High-quality ground-truth (GT) images are crucial for supervised training. We employ



(a) Optical Platform for Low-Light Night Vision



(b) Aperture Lens (c) Integrating Sphere (d) Low-Light Camera

Fig. 8. (a) Optical platform for low-light night vision; (b) collimating objective lens; (c) Integrating sphere; (d) low-light camera.

a hybrid generation strategy tailored to indoor and outdoor conditions.

Indoor scenes: We first capture a reference image at 3×10^{-1} lux using the optical integrating-sphere setup. We then process the reference in Adobe Lightroom by adjusting exposure and contrast, followed by highlights, whites, and blacks, and finally applying noise reduction and sharpening. The operations and parameter settings are listed in Table I.

Outdoor scenes: Because outdoor illumination is spatially nonuniform and cannot be precisely controlled, the indoor workflow cannot be applied directly. We therefore adopt condition-specific procedures for moonlight, starlight, and overcast scenes without direct illumination. For **moonlight scenes** (10^{-1} –1 lux), histogram equalization is first used to increase overall brightness, after which an optical expert adjusts highlights, sharpening, and noise reduction in Lightroom based on the indoor processing protocol. For **starlight scenes** (10^{-2} – 10^{-1} lux), CLAHE is first applied with contrast-limit values in the range 3.0–5.0 and a tile-grid size of 16×16 . An optical expert then adjusts exposure, contrast, highlights, sharpening, and noise reduction for each image. For **overcast scenes without direct illumination**, the illuminance cannot be reliably estimated; these data are typically collected at night under overcast or rainy conditions. Three optical experts jointly inspect and manually adjust these images.

Finally, five optical experts independently review all GT images from both indoor and outdoor settings to reduce

TABLE I
LIGHTROOM OPERATIONS AND SETTINGS

Operation	Function	Range	Guideline Values
Exposure	Increase the overall brightness of the image	1.00–1.50	1.20
Contrast	Enhancing the contrast between the foreground and background	20–35	25
Highlights control	Preventing overexposure in the foreground	-20–10	-15
White color scale	Ensures a more natural-looking white target	20–35	25
Black color scale	Avoid gray shadows	-20–10	-15
Sharpening	Enhance edge sharpness	30–50	40
De-noising	Luminance Noise Reduction	–	Noise Reduction: 40; Detail: 50
	Color Denoising	–	Noise Reduction: 30

subjective bias.

V. EXPERIMENTS

In this section, we detail the experimental setup and present the analysis of the experimental results. We first describe the experimental settings, followed by a comparative evaluation of quantitative and qualitative outcomes. Finally, we conduct ablation studies to analyze the effectiveness and contribution of key modules.

A. Experimental Settings

Implementation Details. Experiments were conducted on a workstation with an Intel Xeon Gold 6330H CPU, 256GB RAM, and an NVIDIA A800 GPU (80GB). The model was implemented in PyTorch 2.4.0. We used the Adam optimizer with an initial learning rate of 1×10^{-4} , which was decayed by a factor of 0.1 every 300 epochs. The batch size was set to 8. Data augmentation included random horizontal and vertical flipping with a probability of 0.4.

Datasets and Evaluation. We evaluate the performance and robustness of MIEFormer on the ELND and LOL datasets. The ELND dataset is divided into indoor and outdoor scenes. For indoor scenes, training and testing are conducted separately for the 1×10^{-3} lux and 3×10^{-4} lux input scenarios; within each scenario, the training and test data have the same input illuminance. The indoor training set contains 8,700 input–target pairs (1,740 input images, each paired with images at five target illumination levels), and the test set contains 300 samples. For outdoor scenes, 900 training samples and 200 test samples were used. During training, the input images were randomly cropped into 256×256 patches. For the 3×10^{-4} lux scenario, the model adopts five ESMs corresponding to the illuminance ranges of $3 \times 10^{-4} \sim 10^{-3}$ lux, $10^{-3} \sim 3 \times 10^{-3}$ lux, $3 \times 10^{-3} \sim 10^{-2}$ lux, $10^{-2} \sim 10^{-1}$ lux, and 10^{-1} lux $\sim$ GT. The number of MFCA modules embedded in each ESM is set to 2, 2, 3, 3, and 2, respectively. For the 10^{-3} lux

scenario, the model employs four ESMs covering the ranges of $10^{-3} \sim 3 \times 10^{-3}$ lux, $3 \times 10^{-3} \sim 10^{-2}$ lux, $10^{-2} \sim 10^{-1}$ lux, and 10^{-1} lux $\sim$ GT, with each ESM containing 2, 3, 3, and 2 MFCA modules, respectively. During inference, the input resolution is fixed to 1280×1024 , and the model outputs only the final prediction. For outdoor settings, it is impractical to accurately measure illuminance levels. Optical experts estimate the minimum illuminance to be 3×10^{-3} lux. Therefore, our model employs three ESMs covering the illuminance ranges $3 \times 10^{-3} \sim 10^{-2}$ lux, $10^{-2} \sim 10^{-1}$ lux, and 10^{-1} lux $\sim$ GT, respectively, with each ESM containing 3, 3, and 2 MFCA modules. Because multi-illumination image sequences are unavailable for outdoor scenes, we retain the feature recovery submodule only in the final ESM. In this setting, the model is trained using only the reconstruction loss and outputs only the final prediction during inference. We use PSNR and SSIM to measure fidelity and LPIPS to assess perceptual similarity.

B. Comparison with Existing Methods

We compare the proposed MIEFormer with a broad set of low-light enhancement methods, including traditional approaches (BPDFHE [49], BPHDE [48], LIME [50]) and deep-learning models (RetinexNet [15], KinD [52], IE-CGAN [51], FiveAPlus [57], ZeroDCE [53], SCI [55], DLN [44], IAT [56], FourLLIE [40], RUAS [54], PIE [59], SwinIR [31], Retinexformer [58], ReDDiT [60], HVI-CIDNet [61], and SG-LLIE [62]). For the ELND dataset, all compared methods are retrained on both indoor and outdoor scenes, and illumination-specific training splits are adopted to ensure a fair comparison. The detailed training configurations are provided in Table III. For the public LOL dataset [15], we retrain only our model, while the officially released pre-trained weights are used for other comparison methods. Since LOL only provides paired low-light and normal-light images rather than multi-illumination sequences, we adapt MIEFormer accordingly by removing the intermediate feature recovery components in all but the final Enhancement Segment Module (ESM). The model is trained only with the reconstruction loss in this setting. The detailed hyperparameter configurations of the traditional comparison methods are provided as follows. For BPDFHE, the blur window size is set to 3, the standard deviation of the Gaussian membership function is set to 1.0762, the local maximum threshold ratio is set to 0.01, the smoothing fusion coefficient is set to 0.2, and the noise suppression weight is set to 0.1. For BPHDE, the number of hierarchical layers is fixed to 2, the maximum grayscale width constraint of hierarchical intervals is set to 6, the stretching factor is set to 2.3, and the fusion coefficient is set to 0.2. For LIME, the standard deviation of the gradient Gaussian kernel is set to 2.0, the kernel window size is set to 15×15 , the regularization weight is set to 0.15, the number of ADMM iterations is set to 10, and the Gamma correction factor is set to 0.8. Based on the empirically estimated minimum illuminance of approximately 10^{-2} lux for LOL images, we configure MIEFormer with two ESMs corresponding to the illumination intervals of $10^{-2} \sim 10^{-1}$ lux and 10^{-1} lux $\sim$ GT, which contain 3 and 2 MFCA blocks, respectively.

TABLE II
QUANTITATIVE RESULTS ON THE LOL DATASET (V1 AND V2-REAL) AND ELND DATASET (INDOOR 1×10^{-3} LUX, INDOOR 3×10^{-4} LUX, OUTDOOR).

Methods	LOL-V1			LOL-V2-real			Indoor (1×10^{-3} lux)			Indoor (3×10^{-4} lux)			Outdoor		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BPDHE [48] [2007]	15.00	0.39	0.48	13.03	0.41	0.48	17.01	0.27	0.65	14.23	0.16	0.74	18.10	0.31	0.65
BPDFHE [49] [2010]	11.38	0.41	0.54	11.96	0.34	0.54	15.89	0.19	0.77	13.12	0.11	0.98	18.25	0.36	0.60
LIME [50] [2016]	15.10	0.48	0.45	15.43	0.41	0.47	14.74	0.18	0.86	13.39	0.13	0.91	18.06	0.29	0.72
IE-CGAN [51] [2017]	16.46	0.51	0.48	15.85	0.53	0.45	12.96	0.10	0.98	12.90	0.10	0.94	16.12	0.20	0.73
RetinexNet [15] [2018]	14.11	0.45	0.38	17.72	0.65	0.44	14.86	0.17	0.74	13.09	0.11	0.97	18.17	0.33	0.62
KinD [52] [2019]	14.87	0.69	0.16	20.01	0.84	0.16	16.78	0.24	0.67	14.68	0.16	0.75	18.29	0.35	0.60
ZeroDCE [53] [2020]	15.35	0.57	0.34	18.06	0.58	0.31	16.19	0.22	0.70	13.58	0.13	0.94	18.16	0.32	0.65
DLN [44] [2020]	20.89	0.72	0.26	20.29	0.75	0.29	17.44	0.31	0.63	15.91	0.19	0.77	18.25	0.34	0.61
RUAS [54] [2021]	23.65	0.52	0.27	15.33	0.49	0.31	17.54	0.32	0.63	13.24	0.12	0.98	18.88	0.38	0.59
SCI [55] [2022]	21.81	0.53	0.36	17.25	0.55	0.32	17.19	0.29	0.71	13.03	0.11	0.98	17.95	0.29	0.71
IAT [56] [2022]	22.80	0.83	0.23	23.13	0.84	0.18	17.47	0.30	0.61	13.53	0.14	0.75	18.57	0.34	0.60
FiveAPlus [57] [2023]	14.73	0.58	0.37	18.16	0.62	0.33	16.02	0.22	0.73	16.32	0.21	0.71	18.31	0.33	0.62
FourLLIE [40] [2023]	22.34	0.85	0.08	22.35	0.85	<u>0.05</u>	17.58	0.30	0.60	15.92	0.19	0.77	18.48	0.36	0.58
Retinexformer [58] [2023]	22.79	0.85	0.13	22.80	0.84	0.17	17.16	0.29	0.64	17.04	0.26	0.66	18.52	0.32	0.63
PIE [59] [2024]	22.54	0.48	0.35	22.04	0.56	0.33	17.69	0.34	0.58	17.12	0.27	0.66	18.36	0.35	0.60
SwinIR [31] [2024]	24.46	<u>0.87</u>	0.07	24.12	0.84	0.17	<u>21.44</u>	<u>0.62</u>	<u>0.44</u>	<u>20.13</u>	<u>0.59</u>	<u>0.51</u>	<u>20.86</u>	<u>0.60</u>	0.47
ReDDiT [60] [2025]	24.29	0.86	<u>0.05</u>	23.92	0.89	0.04	17.39	0.29	0.60	16.16	0.21	0.72	19.01	0.44	0.54
HVI-CIDNet [61] [2025]	25.57	0.89	0.08	<u>24.11</u>	<u>0.87</u>	0.11	18.34	0.46	0.53	17.47	0.34	0.62	20.52	0.48	<u>0.43</u>
SG-LLIE [62] [2025]	<u>24.63</u>	<u>0.87</u>	0.01	22.84	0.86	0.13	17.58	0.33	0.60	17.14	0.26	0.67	19.08	0.46	<u>0.52</u>
Ours	24.33	0.86	0.10	23.98	<u>0.87</u>	0.15	22.49	0.70	0.32	21.29	0.66	0.42	22.68	0.73	0.30

Bold and underlined denote best and second-best performance, respectively. $\uparrow$ indicates higher is better, $\downarrow$ indicates lower is better.

TABLE III
TRAINING SETTINGS OF COMPARED METHODS.

Model	Input Size	Batch Size	Learning Rate	LR Scheduling	Loss Function
RetinexNet	96×96	16	1×10^{-3}	Fixed	L1 + TV smoothness Loss
IE-CGAN	128×128	8	1×10^{-3}	Fixed	L1 + adversarial + VGG perceptual + Sobel edge Loss
KinD	96×96	16	1×10^{-4}	StepLR	L1 + TV smoothness Loss
ZeroDCE	128×128	8	1×10^{-4}	Fixed	Spatial Consistency + Exposure Control + Color Constancy + Illumination Smoothness Loss
DLN	128×128	8	1×10^{-4}	Fixed	L1 + SSIM + TV smoothness Loss
RUAS	128×128	8	1×10^{-4}	Fixed	L2 + TV smoothness Loss
SCI	256×256	4	1×10^{-4}	Fixed	Fidelity Loss + Smoothness Loss
IAT	256×256	4	2×10^{-4}	CosineAnnealingLR	L1 + VGG perceptual Loss
FiveAPlus	256×256	4	4×10^{-4}	CyclicLR	L1 + SSIM + TV smoothness Loss
FourLLIE	256×256	4	4×10^{-4}	MultiStepLR	L2 + Fourier-magnitude TV smoothness Loss
Retinexformer	128×128	8	2×10^{-4}	CosineAnnealingLR	L1 + VGG perceptual Loss
PIE	256×256	4	1×10^{-4}	Fixed	L1 + Contrastive learning Loss
SwinIR	96×96	4	2×10^{-4}	MultiStepLR	Charbonnier + VGG perceptual + TV smoothness Loss
ReDDiT	96×96	4	1×10^{-4}	Fixed	MSE + L1 + TV smoothness Loss
HVI-CIDNet	128×128	8	1×10^{-4}	CosineAnnealingLR	L1 + HVI L1 + HVI Sobel + HVI VGG perceptual Loss
SG-LLIE	128×128	8	2×10^{-4}	Fixed	Charbonnier + VGG perceptual + MSSSIM Loss

Quantitative Comparison. The quantitative results are presented in Table II. On the public LOL dataset [15], deep learning methods consistently outperform traditional techniques across all metrics, with Vision Transformer-based models achieving the highest scores. Our method delivers comparable performance to these leading Transformer-based approaches.

On the proposed ELND dataset, which covers extreme low illumination levels (10^{-4} to 10^{-3} lux), the performance of all compared methods drops noticeably relative to their results on LOL. This indicates that existing methods are primarily tuned for the brighter regime (10^{-1} to 3 lux) and generalize poorly to extremely dark scenes. In contrast, our method maintains high scores across all metrics, demonstrating its robustness and

effectiveness in challenging, near dark night vision conditions.

As listed in Table IV, we quantitatively compare computational complexity (FLOPS) and parameter size (Params). Compared with SwinIR [31] and ReDDiT [60], our approach achieves better joint optimization of parameters and computational efficiency. Furthermore, it avoids the performance degradation of lightweight models such as SCI and RUAS caused by excessive structural simplification and maintains an excellent balance between enhancement quality and computational cost.

Qualitative Comparison. Visual comparisons on the LOL [15] and ELND datasets are shown in Fig. 9. On LOL, differences among methods mainly lie in brightness adjustment

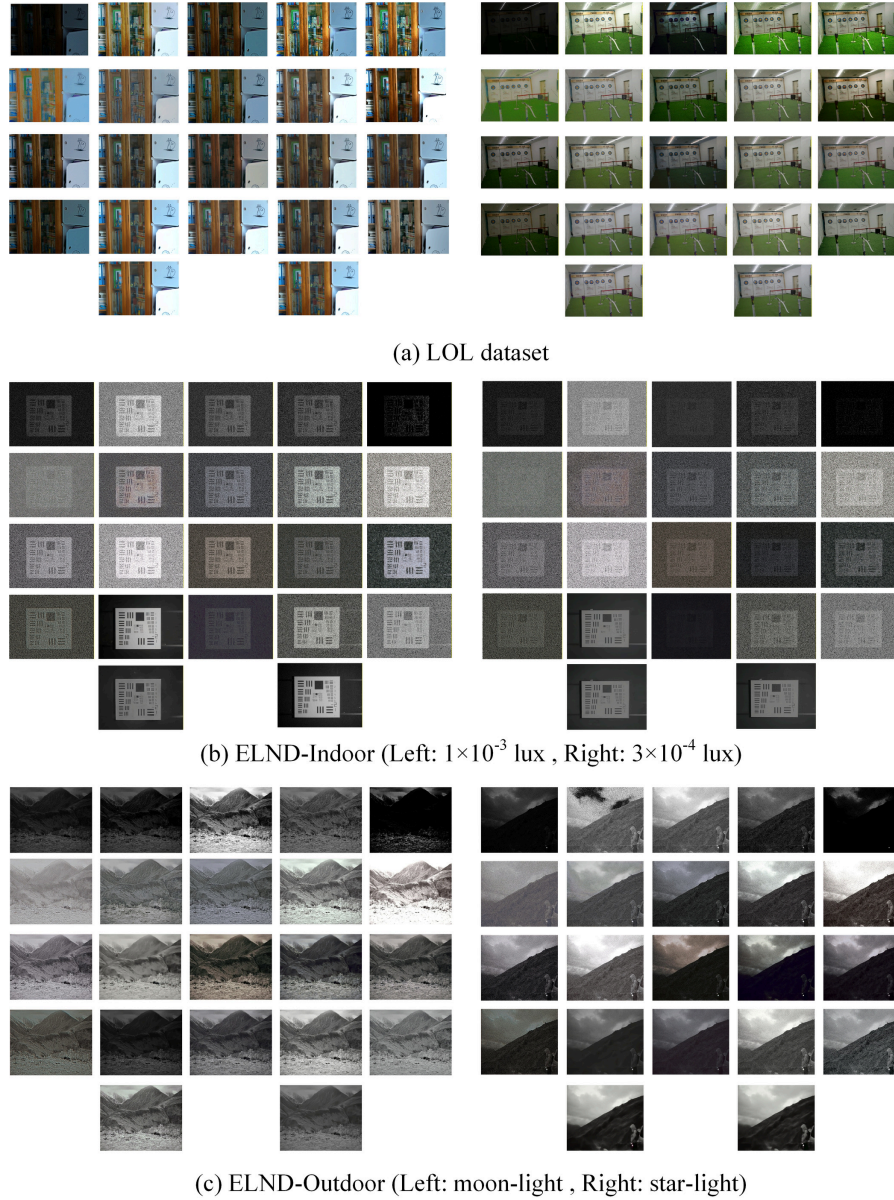


Fig. 9. Comparison of Enhancement Effects in Different Scenarios. Display sequence: 1st row: Input, BPDHE, BPDFHE, LIME, IE-CGAN; 2nd row: RetinexNet, KinD, ZeroDCE, DLN, RUAS; 3rd row: SCI, IAT, FiveAPlus, FourLLIE, Retinexformer; 4th row: PIE, SwinIR, ReDDiT, HVI-CIDNet, SGLLIE; 5th row: Ours, GT. (a) LOL Dataset Image Enhancement Comparison. (b) ELND dataset - Indoor Scene Image Enhancement Comparison (Left: 1×10^{-3} lux, Right: 3×10^{-4} lux). (c) ELND dataset - Outdoor Scene Image Enhancement Comparison (Left: moonlight, Right: starlight).

and color fidelity, where our approach yields more natural and visually consistent results.

C. Hyperparameter Experiments for ESM and MFCA

Based on the piecewise iterative learning design, MIEFormer is decomposed into multiple stages, each equipped with an ESM, and each ESM is further stacked with multiple MFCA blocks. As illustrated in Figures 1(a) and 1(c), the signal-to-noise ratio (SNR) and perceptual differences vary substantially across illumination ranges. For instance, the SNR gap between 3×10^{-4} lux and 10^{-3} lux is smaller than that between 10^{-3} lux and 10^{-2} lux, whereas the gap between 10^{-2} lux and 10^{-1} lux is comparable to the latter in terms of SNR, yet exhibits markedly larger

perceptual differences. The number of ESMs and the depth of MFCA blocks within each ESM both influence overall model performance. To investigate this, we conduct corresponding ablation studies.

The experiments are conducted in controlled indoor scenes at 3×10^{-4} and 10^{-3} lux. We evaluate the design by (1) reducing the number of ESMs and (2) varying the number of MFCA blocks while keeping the ESM count fixed. PSNR and SSIM are used as the evaluation metrics.

Default settings. For the 3×10^{-4} lux scenario, the model comprises 5 stages with 5 ESMs corresponding to the following illumination ranges: ($3 \times 10^{-4} \sim 10^{-3}$ lux), ($10^{-3} \sim 3 \times 10^{-3}$ lux), ($3 \times 10^{-3} \sim 10^{-2}$ lux), ($10^{-2} \sim 10^{-1}$ lux), and (10^{-1} lux-GT). The number of MFCA blocks in

TABLE IV
COMPARISONS OF COMPUTATIONAL COMPLEXITY (FLOPs) AND
PARAMETER COUNT (PARAMS) ACROSS DIFFERENT MODELS.

Methods	FLOPs (G) ↓	Params (M) ↓
IE-CGAN [51] [2017]	790.0	10.50
KinD [52] [2019]	189.0	8.02
ZeroDCE [53] [2020]	120.0	0.079
DLN [44] [2020]	34.81	2.70
RUAS [54] [2021]	75.0	0.45
SCI [55] [2022]	14.9	0.0003
IAT [56] [2022]	75.6	0.09
FiveAPlus [57] [2023]	12.6	0.09
FourLLIE [40] [2023]	102.8	0.13
Retinexformer [58] [2023]	389.0	1.61
PIE [59] [2024]	130.5	0.079
SwinIR [31] [2024]	1120.0	117.20
ReDDiT [60] [2025]	1280.0	66.70
HVI-CIDNet [61] [2025]	1350.0	1.88
SG-LLIE [62] [2025]	1420.0	12.67
Ours	580.0	16.79

each ESM is set to 2, 2, 3, 3, and 2, respectively.

For the 10^{-3} lux scenario, the model comprises 4 stages with 4 ESMs corresponding to: ($10^{-3} \sim 3 \times 10^{-3}$ lux), ($3 \times 10^{-3} \sim 10^{-2}$ lux), ($10^{-2} \sim 10^{-1}$ lux), and (10^{-1} lux~GT). The number of MFCA blocks in each ESM is set to 2, 3, 3, and 2, respectively.

Removing ESM. For the 3×10^{-4} lux scenario, we reduce the number of ESMs from five to four by removing the stage corresponding to $10^{-3} \sim 3 \times 10^{-3}$ lux and expanding the subsequent stage to cover $10^{-3} \sim 10^{-2}$ lux. We apply the same operation to the 10^{-3} lux scenario.

Adjusting the Number of MFCA Blocks. For the 3×10^{-4} lux scenario, based on the default setting, we respectively increase and decrease the number of MFCA blocks within the ESM corresponding to the ($3 \times 10^{-3} \sim 10^{-2}$ lux) stage. The same operations are performed for the 10^{-3} lux scenario under its default setting.

As shown in Table V, reducing the number of ESMs substantially degrades both PSNR and SSIM relative to the default setting, and the degradation is more pronounced at lower illumination. By comparison, changing the number of MFCA blocks within an ESM while keeping the total number of ESMs fixed has a smaller effect. Reducing the number of MFCA blocks degrades the quantitative results, whereas adding blocks provides little or no consistent improvement. The visual results in Fig. 10 support these observations. At 3×10^{-4} lux, reducing the ESM count causes severe quality degradation relative to the default setting. At 10^{-3} lux, the degradation is milder, although reduced brightness and increased noise remain evident. With the ESM count fixed, removing MFCA blocks causes a perceptible decline in quality, whereas adding blocks yields negligible visual improvement. Overall, the visual comparisons in Fig. 10 are consistent with the quantitative trends in Table V.

TABLE V
HYPERPARAMETER EXPERIMENTS FOR ESM AND MFCA

Scenario	Experimental Conditions		PSNR	SSIM
	ESM	MFCA		
Indoor (3×10^{-4} lux)	Default setting	Default setting	21.69	0.66
	Remove ($10^{-3} \sim 3 \times 10^{-3}$ lux)	MFCA: [2,3,3,2]	18.21	0.44
	Default setting	Add number of MFCA	21.67	0.68
	Default setting	Reduce number of MFCA	20.24	0.58
Indoor (10^{-3} lux)	Default setting	Default setting	22.49	0.70
	Remove ($10^{-3} \sim 3 \times 10^{-3}$ lux)	MFCA: [3,3,2]	20.73	0.60
	Default setting	Add number of MFCA	22.51	0.69
	Default setting	Reduce number of MFCA	22.49	0.67

D. Run-Time Testing and Segmentation Validation

Runtime Experiment. Runtime is an important measure of computational efficiency. We compare MIEFormer with representative methods published since 2022. All methods are evaluated on a platform equipped with an Intel Core i7-9750H CPU, 16 GB of RAM, and an NVIDIA RTX 2060 GPU with 6 GB of memory. The test images have a resolution of 1280×1024 and were captured at 1×10^{-3} lux. We report inference time only and use the default configuration of MIEFormer. The results are presented in Table VI.

TABLE VI
QUANTITATIVE RESULTS ON REAL-TIME EXPERIMENT.

Methods	Times (ms) ↓
FiveAPlus [57] [2023]	90
FourLLIE [40] [2023]	190
Retinexformer [58] [2023]	740
PIE [59] [2024]	60
SwinIR [31] [2024]	7800
ReDDiT [60] [2025]	1200
HVI-CIDNet [61] [2025]	230
SG-LLIE [62] [2025]	1030
Ours	320

Considering runtime, FLOPs, and parameter counts, our model shows no inference advantage over lightweight methods, yet achieves lower latency than other non-lightweight counterparts.

Image Segmentation Validation Experiment. We further assess the utility of the proposed method for a downstream image-segmentation task. Specifically, we use Segment Anything as the segmentation model and fix the input resolution and illuminance at 1280×1024 pixels and 3×10^{-4} lux, respectively. As shown in Fig. 11, predictions obtained from our enhanced images preserve finer structures and produce clearer boundaries around the stripes on the target board. These results indicate that MIEFormer can improve the quality of inputs to downstream segmentation models under extremely low illumination.

E. Ablation Studies

To further validate the effectiveness of our design choices, we perform ablation studies to assess the contributions of the proposed piecewise enhancement strategy and its core

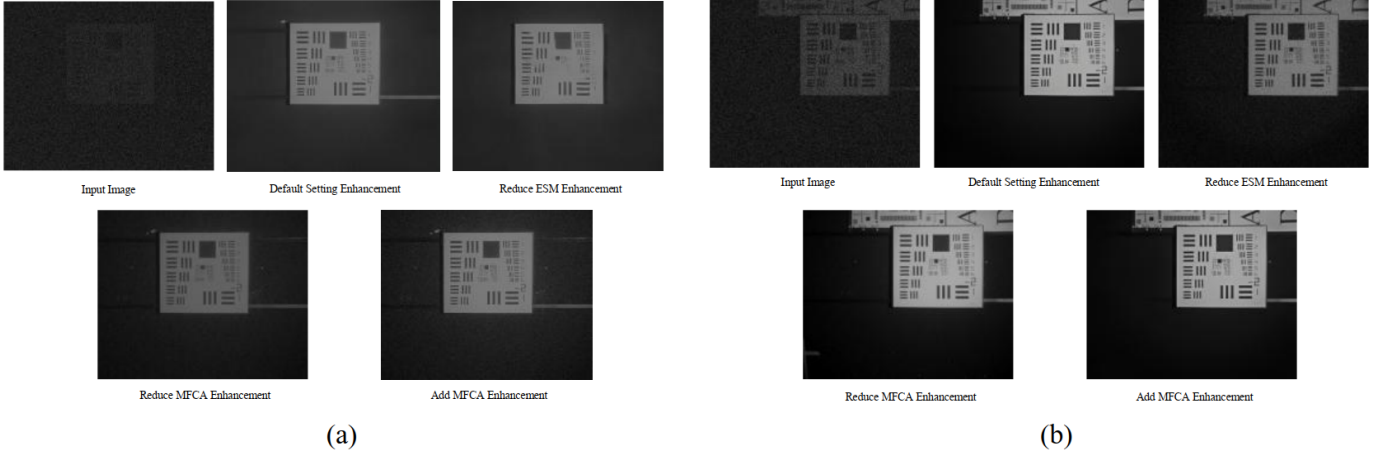


Fig. 10. Results of ablation experiments on the number of ESM and MFCA components; (a) Indoor (3×10^{-4} lux) scenario; (b) Indoor (10^{-3} lux) scenario

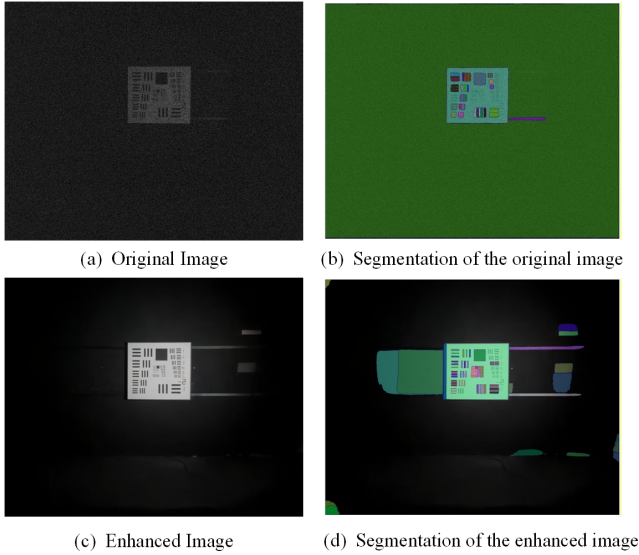


Fig. 11. Comparison of image segmentation results before and after enhancement

modules. All ablations are conducted on the indoor subset of the ELND dataset under an illumination level of 3×10^{-3} lux.

Piecewise Iterative Strategy. MIEFormer adopts a piecewise iterative design where each enhancement stage is supervised separately. This requires training images from multiple illumination levels. Using 3×10^{-3} lux images as input, we compare three supervision schemes: (1) applying loss only at the final output; (2) applying losses at all stages except the 1×10^{-2} lux stage; and (3) applying losses at every stage.

As shown in Fig. 12, the visual results under the three supervision strategies reveal noticeable differences. Throughout the iterative process, gradual improvements in brightness, noise suppression, and detail reconstruction are evident. A key distinction lies in detail preservation: Strategy 3 (full multi-stage supervision) achieves the most accurate detail recovery, whereas Strategy 1 (final output only) and Strategy 2 (partial stage supervision) exhibit progressively greater detail loss. The quantitative results in Tab. VIII align with these

visual observations, confirming that the complete stage-wise supervision strategy most effectively captures and leverages the incremental changes across illumination intervals.

Module Effectiveness. AFBAttention and MCGConv are core components of the MFCA model. To validate the effectiveness of the two modules, we conduct ablation experiments on the same indoor (3×10^{-4} lux) and outdoor test sets adopted in comparative evaluations. Two groups of comparisons are implemented as follows.

For the AFBAttention module, we adjust the number of frequency sub-bands and their corresponding frequency ranges:

- Removing the AFBAttention module;
- Increasing the number of frequency bands to 5;
- Maintaining 4 frequency bands while tuning their start and cutoff frequencies.

For the MCGConv module, we modify its internal convolution components for comparison:

- Removing the MCGConv module;
- Replacing the differential convolution with vanilla convolution;
- Removing only the internal SCDC convolution module.

As shown in Table VII, removing either AFBAttention or MCGConv degrades PSNR and SSIM, with the removal of AFBAttention causing the larger drop. Varying the number of frequency bands or their cutoff frequencies produces only moderate changes in PSNR and SSIM, indicating that AFBAttention is relatively robust to these hyperparameters. Within MCGConv, replacing differential convolutions with standard convolutions also reduces performance, confirming the benefit of the differential structure. Together, these results validate the contributions of both modules.

Temporal Responsibility of the Module. Computational complexity is an important metric for evaluating algorithm performance. Modules with low computational complexity are better suited to the trend toward lightweight models [62]–[65]. To verify the contribution of the proposed MFCA block, we ablate the AFBAttention component by substituting it with three established attention variants: standard Multi-Head Self-Attention (MHSA) [32], Sparse Attention (SAM) [63], and

TABLE VII
ABLATION STUDY ON THE EFFECTIVENESS OF ABFATTENTION AND MCGCONV.

Dataset	Metric	w/o ABFAtt.	Bands=5	Bands=4	w/o MCGConv	w/o DiffConv	w/o SCDC
Indoor (10^{-3} lux)	PSNR	17.36	22.52	22.44	21.19	21.39	22.23
	SSIM	0.32	0.72	0.69	0.60	0.61	0.67
Indoor (3×10^{-4} lux)	PSNR	16.56	21.32	21.21	20.02	20.77	21.03
	SSIM	0.23	0.68	0.64	0.58	0.62	0.64
Outdoor	PSNR	18.37	22.69	22.64	20.41	21.56	22.48
	SSIM	0.35	0.71	0.69	0.63	0.66	0.69

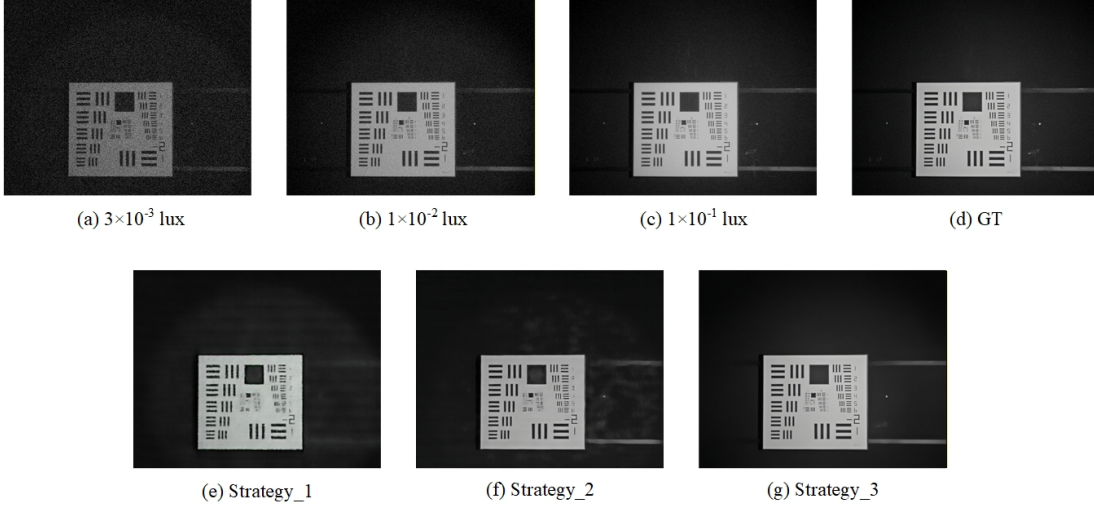


Fig. 12. Low-light night vision images under different illuminance conditions and enhanced images using three distinct strategies.

TABLE VIII
COMPARISON EXPERIMENTS OF THREE DIFFERENT STUDY STRATEGIES.

Strategy	Illumination Levels			PSNR $\uparrow$	SSIM $\uparrow$
	1×10^{-2} lux	1×10^{-1} lux	GT		
Strategy 1	$\times$	$\times$	$\checkmark$	20.89	0.62
Strategy 2	$\times$	$\checkmark$	$\checkmark$	21.16	0.63
Strategy 3	$\checkmark$	$\checkmark$	$\checkmark$	22.86	0.69

TABLE IX
COMPARATIVE EXPERIMENT OF ATTENTION MODULES.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	Time Complexity
MHSA	22.67	0.61	$\mathcal{O}((H \times W)^2 d_{\text{model}})$
EAM	20.81	0.62	$\mathcal{O}((H \times W) d_{\text{model}} S)$
SAM	21.04	<u>0.63</u>	$\mathcal{O}((H \times W) d_{\text{model}} \log(H \times W))$
Ours	<u>21.85</u>	0.65	$\mathcal{O}((H \times W) d_{\text{model}} \log(H \times W))$

Bold and underlined denote best and second-best performance, respectively. H , W , d_{model} , and S denote the height, width, channel dimension, and size of the external memory, respectively.

External Attention (EAM) [64]. The quantitative results of this comparison are summarized in Table IX.

In terms of time complexity, the standard multi-head self-attention module incurs the highest cost at $\mathcal{O}((H \times W)^2 d_{\text{model}})$, while the proposed ABFAAttention

module achieves a substantially lower complexity of $\mathcal{O}((H \times W) d_{\text{model}} \log(H \times W))$. Other efficient variants like SAM [63] and EAM [64] also operate at reduced complexity. Visually (Fig. 13), the MHSA-based version produces overall overexposure, with noticeable jagged edges and blurred or lost details in central regions. While SAM [63] and EAM [64] avoid global overexposure, EAM [64] yields extensively blurred details, and SAM [63] leads to significant detail loss despite avoiding local blurring. In contrast, ABFAAttention maintains better detail integrity and exposure balance. These comparisons demonstrate that our ABFAAttention module effectively preserves visual quality while offering lower computational complexity.

VI. CONCLUSION

In this paper, we addressed a key limitation in current low-light enhancement research: poor adaptability to extremely low-illumination night-vision conditions (below 10^{-3} lux). To bridge the critical data gap in this domain, we first constructed the Enhanced Low-Light Night Vision Dataset (ELND), a multi-scene benchmark covering indoor illuminance ranging from 1×10^{-1} to 3×10^{-4} lux, along with outdoor scenes under moonlight, starlight, and complete darkness. We then proposed a novel piecewise iterative enhancement framework, termed MIEFormer, which decomposes the illumination spectrum into discrete frequency bands and progressively refines features through successive residual learning stages. Experiments

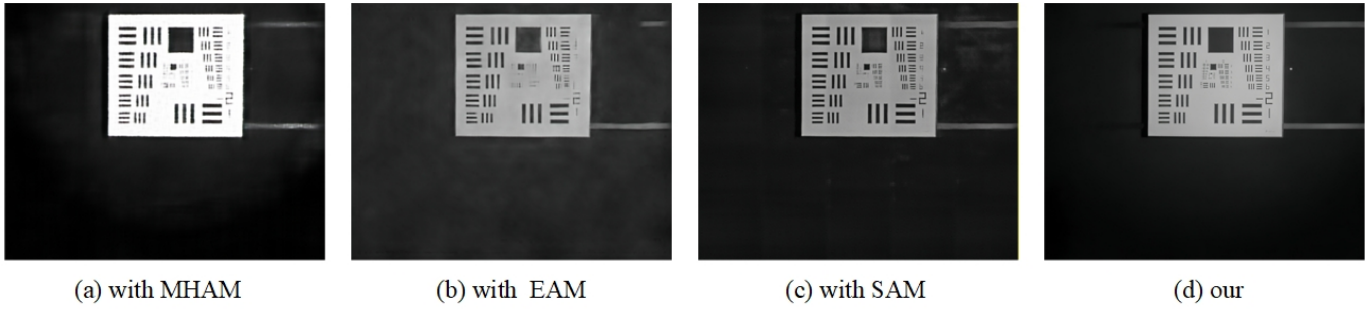


Fig. 13. Enhanced images using different attention modules.

conducted on both the public LOL dataset and our ELND benchmark demonstrate that MIEFormer achieves state-of-the-art performance in terms of quantitative metrics and visual quality, confirming its effectiveness and generalization capability. Despite these promising results, MIEFormer remains sensitive to challenging degradations that commonly occur in night-vision imagery, including severe motion blur, abrupt spatial illumination variations, and ambiguous foreground-background boundaries. These factors can reduce enhancement quality and reliability. Future work will therefore focus on improving robustness to such degradations and optimizing the architecture for real-time deployment.

ACKNOWLEDGMENTS

This work was supported in part by the grants from the National Natural Science Foundation of China (No.62406264, 61727809).

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